Machine Learning 1

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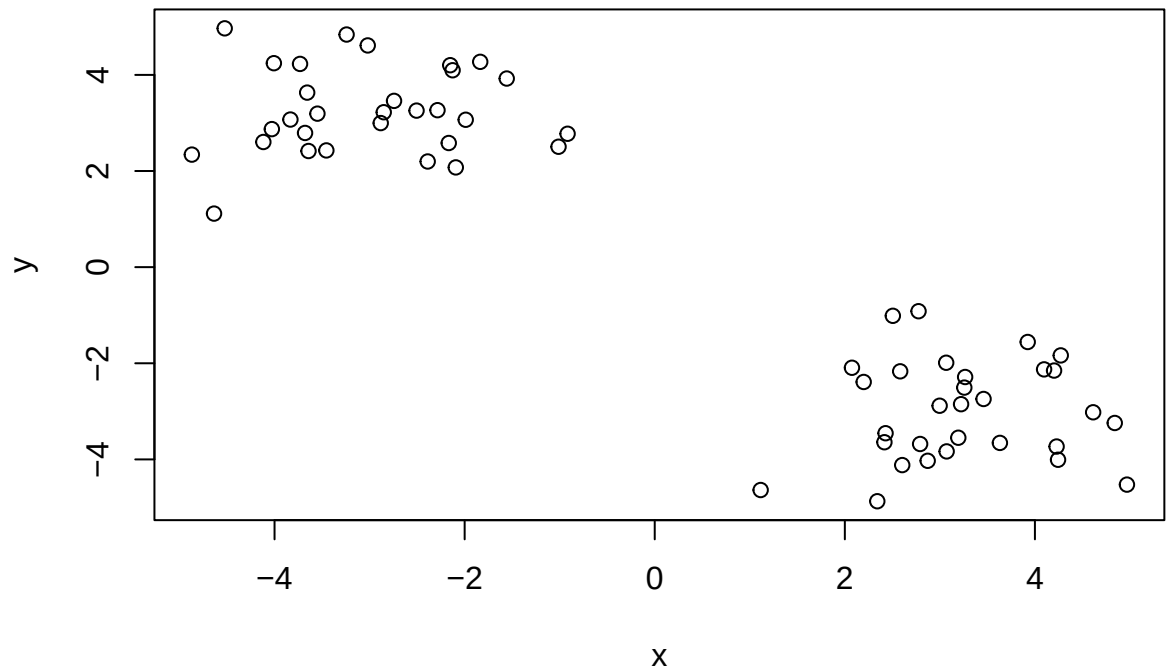
First up kmeans()

Demo of using kmeans() function in based R. First make up some data with a known structure.

```
tmp <- c( rnorm(30, -3), rnorm(30, 3) )
tmp
```

```
##  [1] -2.742862 -4.028536 -3.019539 -3.679115 -2.505984 -2.167190 -4.117917
##  [8] -3.548893 -3.454849 -1.557146 -1.835944 -1.012101 -2.882948 -1.988668
## [15] -2.388784 -2.126683 -4.870197 -3.833257 -2.092999 -4.636551 -4.007214
## [22] -2.851073 -0.916853 -4.524433 -3.731743 -2.151513 -3.656666 -3.642700
## [29] -3.242113 -2.285577  3.265760  4.839859  2.416428  3.631603  4.201469
## [36]  4.228132  4.968279  2.775532  3.223182  4.243322  1.114390  2.074945
## [43]  3.071195  2.341631  4.096453  2.198507  3.066428  2.997500  2.505219
## [50]  4.271693  3.924154  2.428444  3.193883  2.603805  2.582573  3.256925
## [57]  2.792146  4.612630  2.871850  3.459643
```

```
x <- cbind(x=tmp, y=rev(tmp))
plot(x)
```



Now we have some made up data in `x` let's see how kmeans works with this data

```
k <- kmeans(x, centers = 2, nstart = 20)
k
```

```
## K-means clustering with 2 clusters of sizes 30, 30
##
## Cluster means:
##           x         y
## 1 -2.983335  3.241919
## 2  3.241919 -2.983335
##
## Clustering vector:
##  [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2
## [39] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##
## Within cluster sum of squares by cluster:
## [1] 55.85505 55.85505
##  (between_SS / total_SS =  91.2 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

Q. How many points are in each cluster?

```r
k$size
```

```
## [1] 30 30
```

Q. How do we get to the cluster membership/assignment.

```r
k$cluster
```

```
##  [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2
## [39] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
```

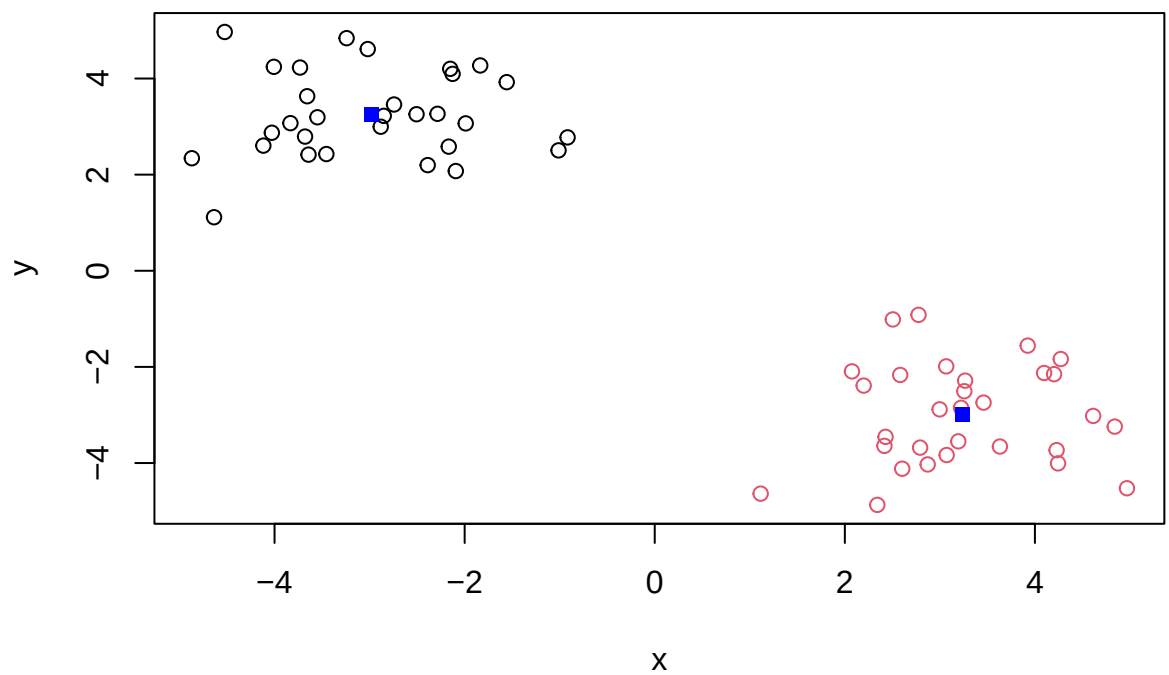
Q. What about cluster centers?

```r
k$centers
```

```
##           x         y
## 1 -2.983335  3.241919
## 2  3.241919 -2.983335
```

Now we got to the main results. Let's use them to plot our data with the kmeans result

```r
plot (x, col=k$cluster)
points(k$centers, col="blue", pch=15)
```



Now for hclust()

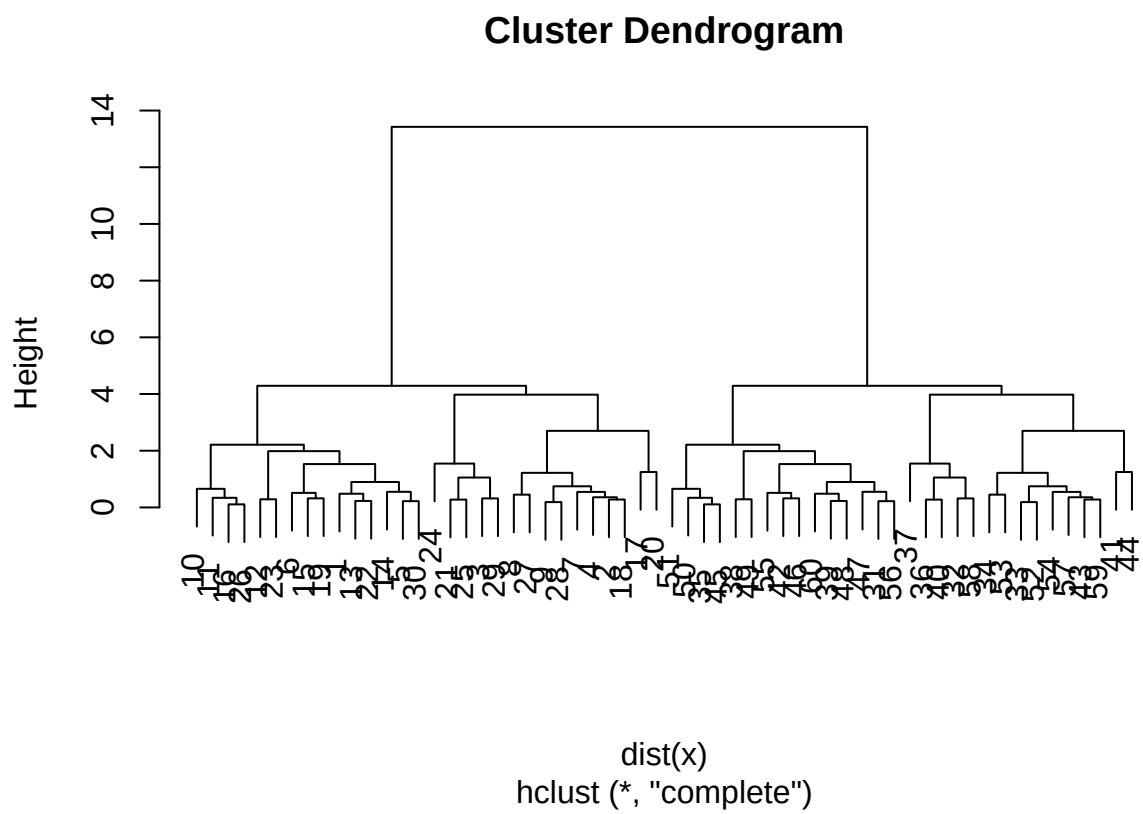
We will cluster the same data `x` with the `hclust()`. In this case `hclust()` requires a distance matrix as input.

```r
hc <- hclust( dist(x) )
hc
```

```
##
## Call:
## hclust(d = dist(x))
##
## Cluster method   : complete
## Distance         : euclidean
## Number of objects: 60
```

Let's plot our hclust result

```r
plot(hc)
```



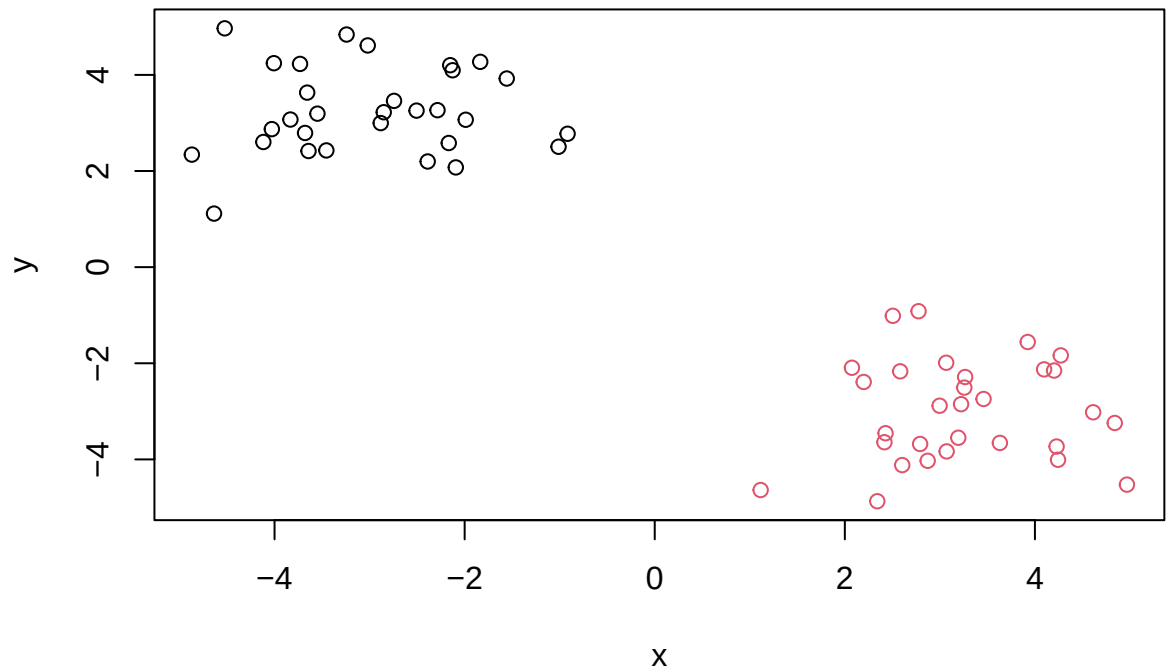
To get our cluster membership vector we need to "cut" the tree with the `cutree()`

```r
grps <- cutree(hc, h=8)
grps
```

```
##  [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2
## [39] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
```

Now plot our data with the hclust() results.

```r
plot(x, col=grps)
```



Principal Component Analysis (PCA)

PCA of UK food data

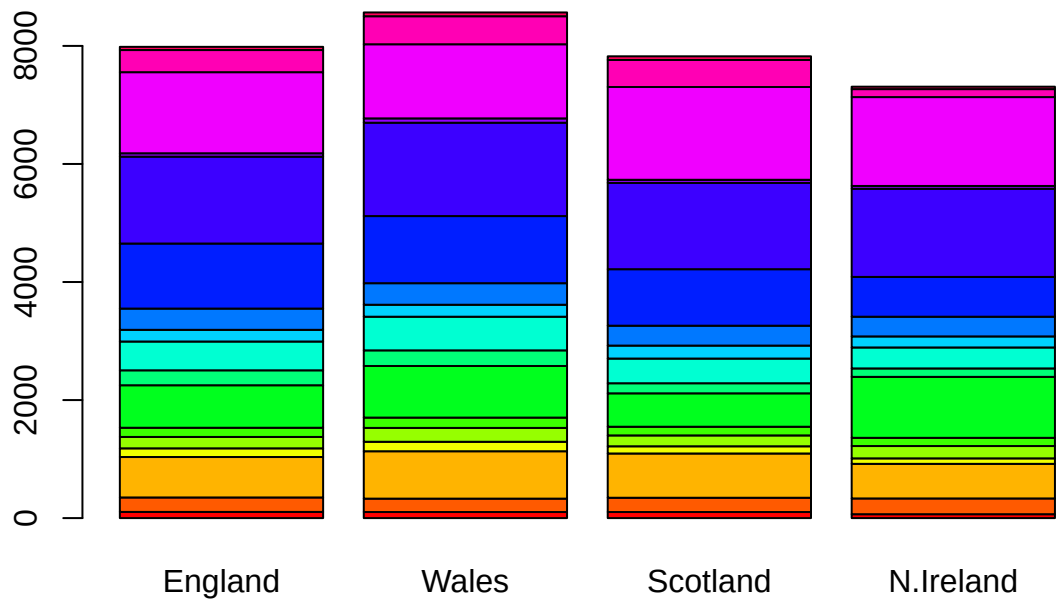
Read data from website and try a few visualizations.

```
url <- "https://tinyurl.com/UK-foods"
x <- read.csv(url, row.names=1)
x
```

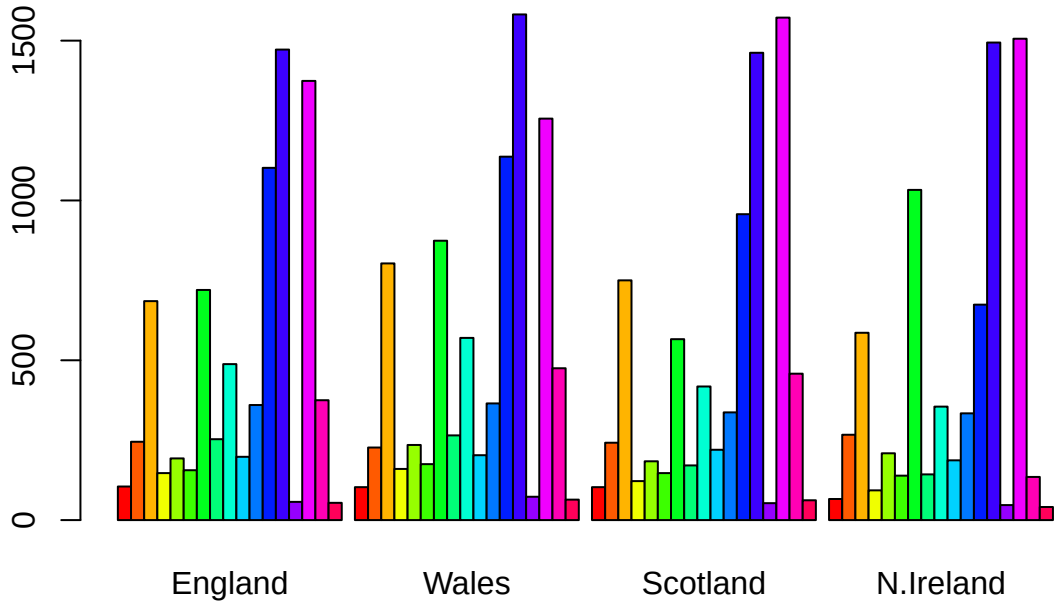
```
##                     England Wales Scotland N.Ireland
## Cheese                  105   103      103        66
## Carcass_meat            245   227      242       267
## Other_meat              685   803      750       586
## Fish                    147   160      122        93
## Fats_and_oils           193   235      184       209
## Sugars                  156   175      147       139
## Fresh_potatoes          720   874      566      1033
## Fresh_Veg               253   265      171       143
## Other_Veg               488   570      418       355
## Processed_potatoes      198   203      220       187
## Processed_Veg           360   365      337       334
## Fresh_fruit            1102  1137      957       674
## Cereals                1472  1582     1462      1494
## Beverages                57    73       53        47
## Soft_drinks            1374  1256     1572      1506
```

```
## Alcoholic_drinks        375   475      458       135
## Confectionery            54    64       62        41
```

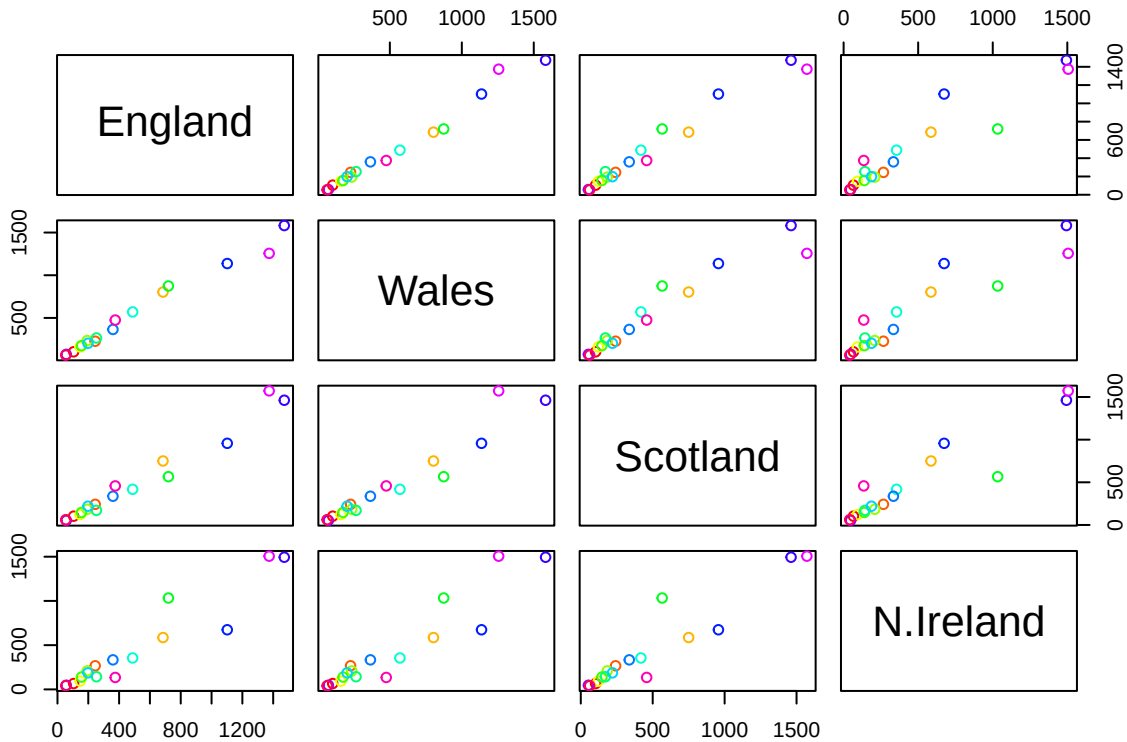
```r
cols <- rainbow(nrow(x))
barplot( as.matrix(x), col=cols )
```




```r
barplot( as.matrix(x), col=cols, beside = TRUE )
```




```r
pairs(x, col=cols)
```



PCA to the rescue!! The main base R PCA function is called `prcomp()` and we will ened to give it the transpose of our input data!

```
t(x)
```

```
##           Cheese Carcass_meat  Other_meat  Fish Fats_and_oils  Sugars
## England      105           245         685  147            193     156
## Wales        103           227         803  160            235     175
## Scotland     103           242         750  122            184     147
## N.Ireland     66           267         586   93            209     139
##           Fresh_potatoes  Fresh_Veg  Other_Veg  Processed_potatoes
## England               720        253        488                 198
## Wales                 874        265        570                 203
## Scotland              566        171        418                 220
## N.Ireland            1033        143        355                 187
##           Processed_Veg  Fresh_fruit  Cereals  Beverages Soft_drinks
## England               360         1102     1472         57         1374
## Wales                 365         1137     1582         73         1256
## Scotland              337          957     1462         53         1572
## N.Ireland             334          674     1494         47         1506
##           Alcoholic_drinks  Confectionery
## England                375             54
## Wales                  475             64
## Scotland               458             62
## N.Ireland              135             41
```

```r
pca <- prcomp( t(x) )
```

```r
summary(pca)
```

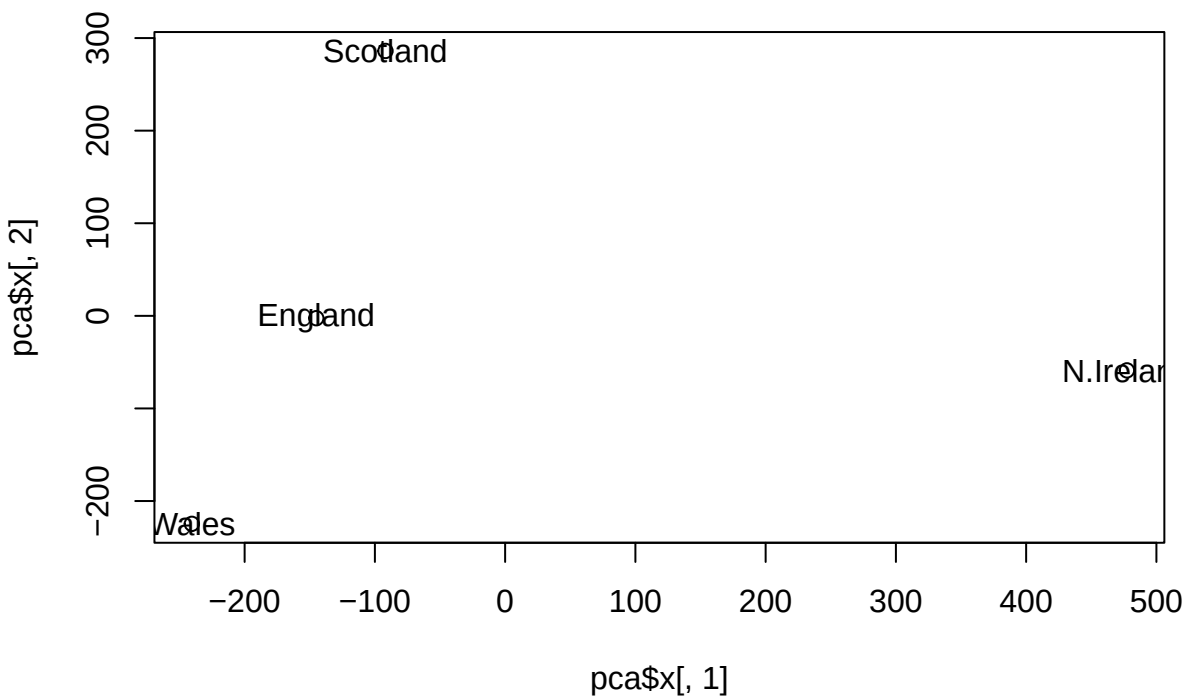
```
## Importance of components:
##                              PC1      PC2      PC3     PC4
## Standard deviation     324.1502 212.7478 73.87622 2.7e-14
## Proportion of Variance   0.6744   0.2905  0.03503 0.0e+00
## Cumulative Proportion    0.6744   0.9650  1.00000 1.0e+00
```

```r
attributes(pca)
```

```
## $names
## [1] "sdev"     "rotation" "center"   "scale"    "x"
##
## $class
## [1] "prcomp"
```

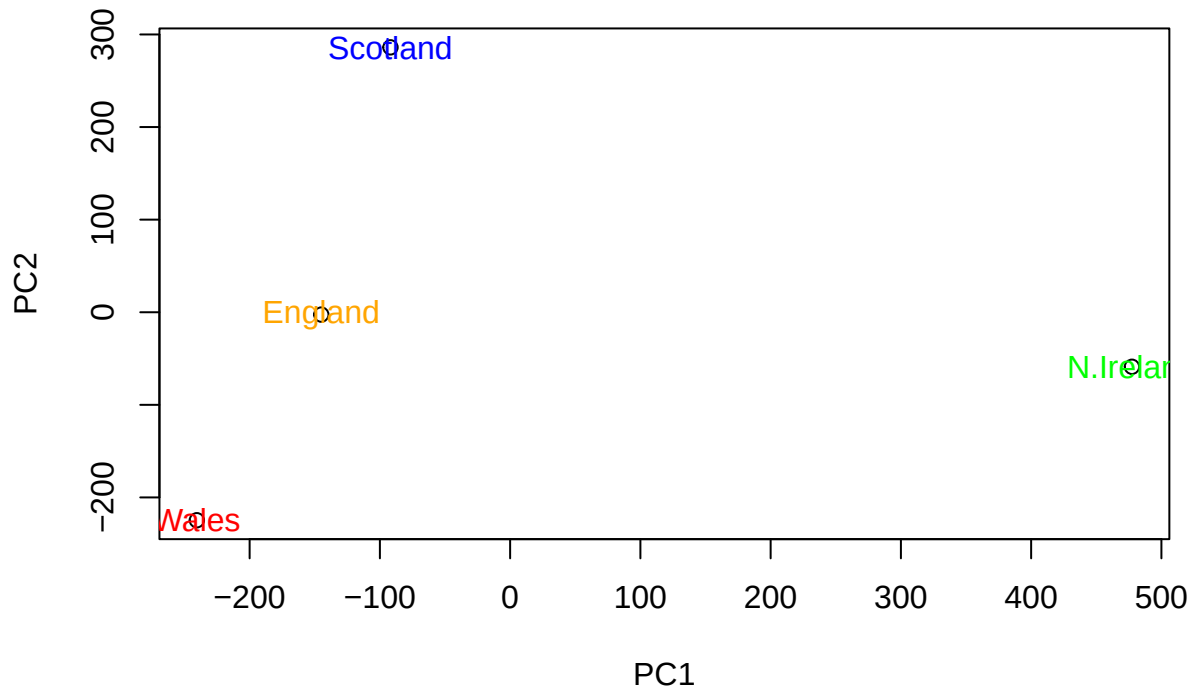
To make our new PCA plot (a.k.a. PCA score plot) we access `pca$x`

```r
plot(pca$x[,1], pca$x[,2])
text(pca$x[,1], pca$x[,2], colnames(x))
```



Color up the plot

```r
country_cols <- c("orange", "red", "blue", "green")
plot(pca$x[,1], pca$x[,2], xlab="PC1", ylab="PC2")
text(pca$x[,1], pca$x[,2], colnames(x), col=country_cols)
```



PCA of RNA-Seq data

Read in data from website

```
url2 <- "https://tinyurl.com/expression-CSV"
rna.data <- read.csv(url2, row.names=1)
head(rna.data)
```

```
##        wt1 wt2  wt3  wt4 wt5 ko1 ko2 ko3 ko4 ko5
## gene1  439 458  408  429 420  90  88  86  90  93
## gene2  219 200  204  210 187 427 423 434 433 426
## gene3 1006 989 1030 1017 973 252 237 238 226 210
## gene4  783 792  829  856 760 849 856 835 885 894
## gene5  181 249  204  244 225 277 305 272 270 279
## gene6  460 502  491  491 493 612 594 577 618 638
```

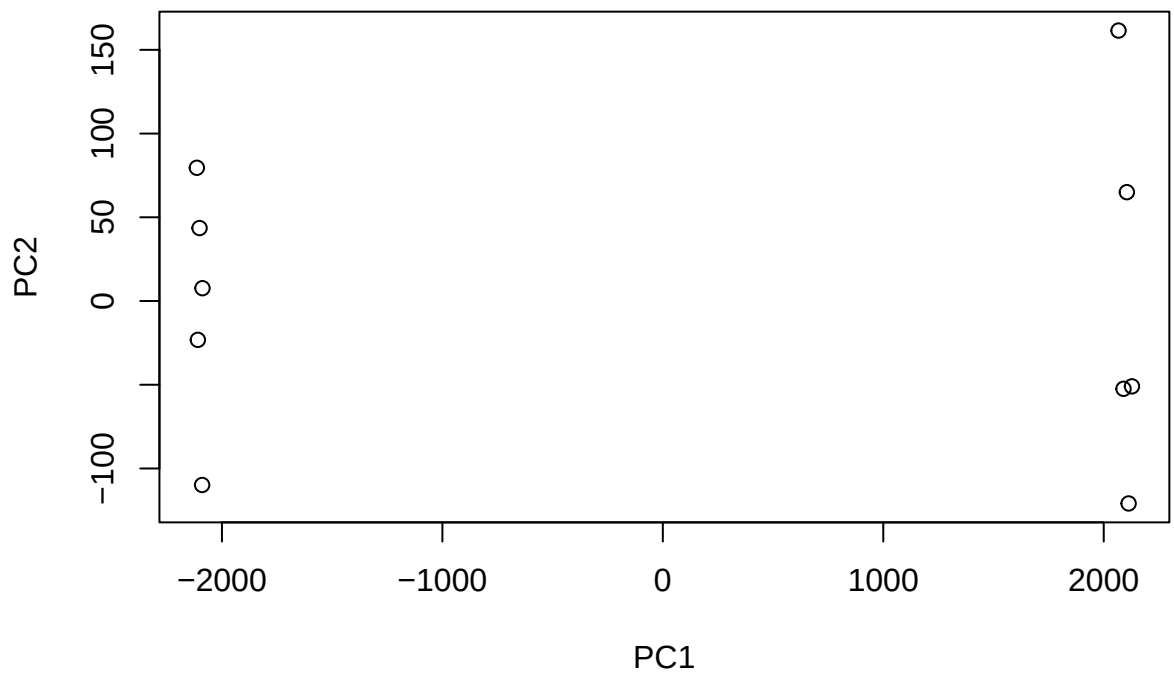
```
pca <- prcomp( t(rna.data) )
summary(pca)
```

```
## Importance of components:
##                             PC1     PC2      PC3      PC4      PC5      PC6
## Standard deviation     2214.2633 88.9209 84.33908 77.74094 69.66341 67.78516
## Proportion of Variance    0.9917  0.0016  0.00144  0.00122  0.00098  0.00093
## Cumulative Proportion     0.9917  0.9933  0.99471  0.99593  0.99691  0.99784
##                            PC7      PC8      PC9      PC10
## Standard deviation     65.29428 59.90981 53.20803 2.852e-13
## Proportion of Variance  0.00086  0.00073  0.00057 0.000e+00
```

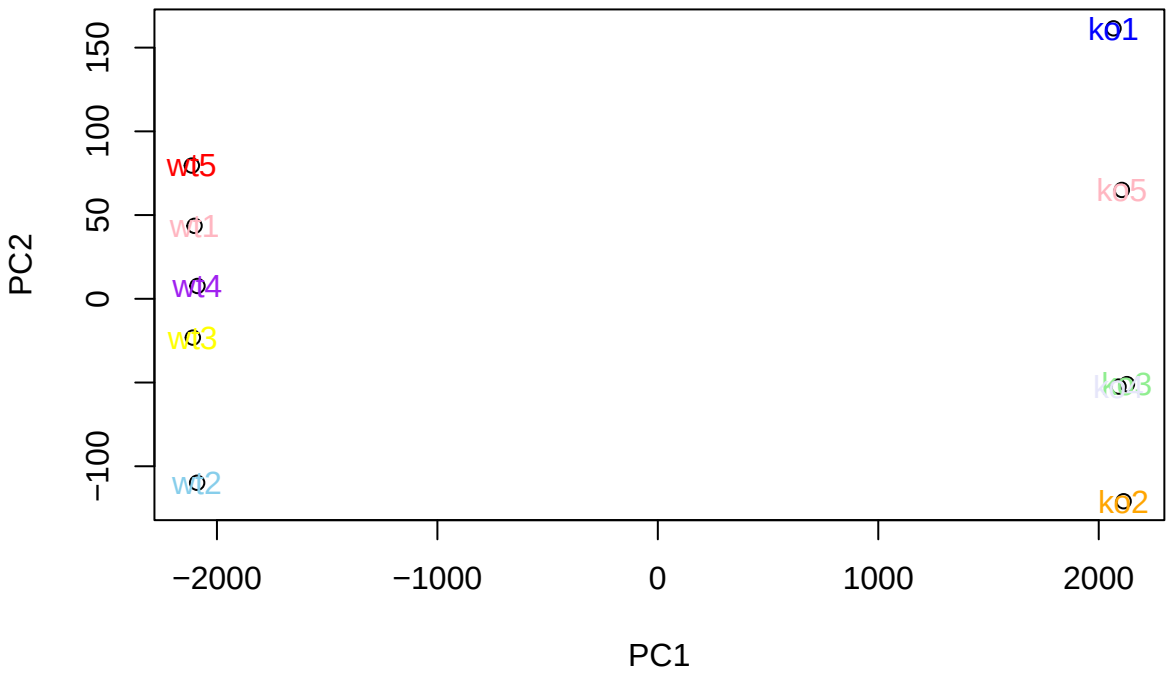
```
## Cumulative Proportion   0.99870  0.99943  1.00000 1.000e+00
```

Do our PCA plot of this RNA-Seq data

```
plot(pca$x[,1], pca$x[,2], xlab="PC1", ylab="PC2")
```




```
gene_cols <- c("lightpink", "skyblue", "yellow", "purple", "red", "blue", "orange", "lightgreen", "laven
plot(pca$x[,1], pca$x[,2], xlab="PC1", ylab="PC2")
text(pca$x[,1], pca$x[,2], colnames(rna.data), col=gene_cols)
```


ko1
ko5
ko3
ko2
wt5
wt1
wt4
wt3
wt2
150
100
50
0
−100
PC2
−2000
−1000
0
1000
2000
PC1